

ORIGINAL ARTICLE

Tokishakuyakusan, a traditional Japanese medicine (Kampo) mitigates iNKT cell-mediated pregnancy loss in mice

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Objective: Tokishakuyakusan (TSS) is a traditional herbal medicine that has been used empirically to prevent recurrent pregnancy loss. Its mode of action remains unclear. With their potent capacity to produce cytokines, invariant natural killer (iNKT) cells are involved in the control of fetomaternal immunity in early gestation. This study aimed to clarify the effect of TSS on iNKT cell activities in a well-studied murine miscarriage model.

Methods: Pregnant mice were fed 1% TSS-containing or control diet from the day of vaginal plug formation. Alpha-galactosylceramide (AGC) was administered intraperitoneally to the pregnant mice at day 9.5 postcoitus (pc) to stimulate iNKT cells. Peripheral cytokine levels were evaluated using cytokine arrays. The percentage of iNKT cells among splenocytes was examined by flow cytometric analysis. The incidence of pregnancy loss was assessed at day 12.5 pc.

Results: The ratio of fetal resorptions to total conceptuses was significantly higher in the group exposed to TSS (34%) than in controls (78%). A rapid and robust surge in inflammatory cytokines, including IFN- γ and TNF- α , was detected in the peripheral blood of control animals 2 hours after AGC administration. This peripheral cytokine induction was significantly attenuated in the TSS-fed group compared with the control. The percentage of iNKT cells among total splenocytes was lower in the TSS-fed group than in controls.

Conclusion: The findings in this study suggest that the inhibitory effects of TSS on pregnancy loss may involve immune modulation of iNKT cells during early pregnancy.

KEYWORDS

invariant natural killer cells, miscarriage, pregnancy, Tokishakuyakusan

1 | INTRODUCTION

Elaborate tuning of the maternal immune system is required to harmonize fetomaternal interactions during pregnancy. The expansion of specialized T-cell subsets with immunosuppressive properties

(eg CD4⁺Foxp3⁺ regulatory T cells¹) and the establishment of a pro-tolerogenic cytokine environment^{2,3} have been reported to be critical to maternal receptivity of a semi-allogeneic fetus in early pregnancy. Impairments in the modulation of maternal immune responses have been associated with multiple adverse pregnancy outcomes, including recurrent pregnancy loss (RPL), fetal growth restriction and pre-eclampsia.^{4,5}

Takeshi Nagamatsu and Tatsuya Fujii equally contributed to this study.

Clinical practice in Japan uses a broad array of diagnostic and treatment approaches that include the use of herbal medications that are a routine part of strategies that fall under the auspices of traditional “Kampo medicine.” Prescription of Japanese herbal medications is authorized by the Ministry of Health, Labor and Welfare of Japan. Tokishakuyakusan (TSS) is a representative Kampo medicine formula that has been widely and empirically used for recurrent miscarriage and unexplained fetal growth restriction. TSS is composed of six herbal ingredients: *Paeoniae Radix*, *Atractylodis Lanceae Rhizoma*, *Alismatis Rhizoma*, *Hoelen*, *Cnidii Rhizoma* and *Angelicae Radix*, which are currently manufactured under strict quality controls with Japan's good manufacturing practice (GMP). Our understanding of the pharmacological mechanisms underlying the effects of this commonly used therapeutic is regrettably limited, particularly regarding their purported effects on pregnancy maintenance and health. Our group previously reported that TSS modulates the balance of Th1 and Th2 cytokine production in human decidual mononuclear cells.⁶ Granulocyte-macrophage colony-stimulating factor (GM-CSF) had been demonstrated as a key growth factor that facilitates embryo development in the peri-implantation period, and we further demonstrated that TSS promoted the production of GM-CSF in decidual stromal cells.⁷

Invariant natural killer T (iNKT) cells are a minor subset of lymphocytes comprising just 0.1%–0.01% of peripheral mononuclear cells. iNKT cells are characterized by their restricted repertoire of T-cell receptor (TCR) alpha chains, V α 14-J α 18 in mice and V α 24-J α 18 in humans.⁸ Using these unique TCRs, iNKT cells recognize glycolipids present on CD1d, a monomorphic class I MHC-like molecule.⁹ Upon activation through the recognition of endogenous or exogenous glycolipid antigens, iNKT cells rapidly release a wide array of cytokines.⁸ Accumulating evidence in mice and humans suggests significant involvement of iNKT cells in the determination of fetomaternal immunity in early pregnancy.^{10–13} Previous works have demonstrated that excessive activation of iNKT cells by alpha-galactosylceramide (AGC), a specific CD1d ligand, induces pregnancy loss in mice.^{10,12} In addition, J α 18^{−/−} mice deficient in iNKT cells were shown to be resistant to lipopolysaccharide-induced preterm birth and fetal death.¹⁴

This study aimed to investigate effects of TSS on pregnancy loss and their relevance to the biological function of iNKT cells. We studied the impact of TSS administration on AGC-induced pregnancy loss, the modulation of cytokine release from AGC-stimulated iNKT cells and the frequency of iNKT cells residing in the spleen following TSS administration in a murine model of iNKT cell-mediated abortion.

2 | MATERIALS AND METHODS

2.1 | The mouse model

All experiments were performed under the approval of the Institutional Animal Care and Use Committee of the University of Tokyo. C57BL/6 J mice were purchased from Japan SLC, Inc. (Hamamatsu, Shizuoka, Japan). Female mice were mated at 9 to

12 weeks of age. Day 0.5 postcoitus (pc) was defined as the morning of vaginal plug detection, and mice with a vaginal plug were randomly divided into two groups at this time point. There were no significant differences between the two groups in terms of age, body weight, pregnancy rate and number of embryos per pregnancy. The animals were allowed free access to water and food, standard laboratory food. From day 0.5 pc until the day of the following assessment, the intervention group of mice was fed a diet containing 1% TSS (Tsumura & Co, Tokyo, Japan) and the other group (control group) was given the standard laboratory food (Labo MR Stock; Nosan Corporation, Yokohama, Japan) without TSS. In the group given TSS-containing food, 30–40 mg/body/d was estimated as the average TSS intake amount based on the food consumption. On day 9.5 pc, AGC (Kirin Co Ltd., Tokyo, Japan) was administered intraperitoneally at a dose of 40 μ g/kg to all animals. Tail vein blood was sampled immediately before and 2 hours after AGC administration. Mice were killed 72 hours after AGC administration, and the rates of the fetal resorption and fetoplacental weights were determined. “Resorption rate” was defined as the number of resorbed fetuses divided by the total number of the fetuses in the uterus for each mouse multiplied by 100%. Blood was also collected from the hearts of the dams at the time of killing.

2.2 | Histological assessments

The uteri were removed from the pregnant mice at day 12.5 pc. Fetoplacental tissues were fixed in 4% paraformaldehyde-phosphate-buffered solution (Wako, Osaka, Japan) for 24 hours. Samples were embedded in paraffin after fixation and then dissected at a thickness of 6 μ m. Deparaffinized sections were stained with Gill's hematoxylin and eosin.

2.3 | Cytokine arrays

Blood samples were obtained from the pregnant mice at 0, 2 and 72 hours after AGC stimulation and were left to stand at room temperature until clotting was completed. The samples were then centrifuged at 2000 g for 20 minutes, and serum was collected. Serum samples were stored at -80°C until use for batched cytokine measurement. Serum cytokine levels for IL-2, IL-4, IL-6, IL-10, IL-13, IL-17, interferon (IFN)- γ , tumor necrosis factor (TNF)- α , IL-12 (p70), GM-CSF and monocyte chemoattractant protein (MCP)-1 were measured using the Bio-Plex Suspension Array System (Bio-Rad Laboratories, Hercules, CA, USA) and a MILLIPLEX MAP Mouse Cytokine/Chemokine Magnetic Bead Panel (Millipore, Billerica, MA, USA) according to the manufacturer's instructions. A standard curve specific to each cytokine was generated by applying standard solutions with known cytokine concentrations on the same array panel to avoid inter-lot variation errors.

2.4 | Isolation of splenocytes

At day 7.5 pc and day 9.5 pc, the spleens were removed from killed mice. Spleens were incised on a sterile dish containing

phosphate-buffered saline (PBS), and the dissociated splenocytes were collected in this buffer. After washing once with PBS, the splenocytes were incubated in 10 mL of RBC lysis buffer (BD Biosciences, San Jose, CA, USA) for 10 minutes at 4°C. The resulting cell suspension was centrifuged at 800 *g* for 5 minutes and then washed twice with PBS. The isolated splenocytes were immediately analyzed by flow cytometry.

2.5 | Flow cytometric analysis

Flow cytometric analysis was used to detect iNKT cells. Splenocytes were incubated with Purified Rat Anti-Mouse CD16/CD32 (BD Biosciences) for 10 minutes at 4°C to block Fc receptor-mediated reactions. Splenocytes were then incubated with fluorochrome-conjugated anti-mice antibodies for 30 minutes at 4°C in the dark. A combination of PE Hamster Anti-Mouse TCR β Chain (BD Biosciences) and APC T-Select Mouse CD1d Tetramer loaded with AGC (MBL, Nagoya, Japan) was used to identify iNKT cells. Empty CD1d tetramer without AGC loading was used for a negative control to check the specific binding of AGC-CD1d tetramer to iNKT cells. Following antibody exposure, the splenocyte suspensions were washed twice and resuspended in a volume of 0.5 mL staining buffer. A cell subset corresponding to total splenic lymphocytes was identified on side scatter to forward scatter plot. Gating to the total lymphocyte subset, further analysis to detect iNKT cells was conducted. Double-positive (TCR β^+ , CD1d tetramer $^+$) cells were identified as iNKT cells.

2.6 | Statistical analysis

Differences in serum cytokine levels at each time point were evaluated by analysis of variance followed by a post hoc analysis

using the Dunnett's test. Wilcoxon rank sum test was used to compare pregnancy outcomes, serum cytokine levels and cell subset distributions between the TSS-fed group and the control group. $P < 0.05$ was considered statistically significant. All the statistical analyses were performed using JMP Pro 11 software (SAS Institute, Cary, NC, USA).

3 | RESULTS

3.1 | Impact of TSS feeding on AGC-induced fetal resorption

The incidence of pregnancy loss at 72 hours after iNKT cell activation by AGC administration was examined in the TSS-fed and control groups. Representative findings of AGC-induced pregnancy loss in this model are shown in Figure 1A,B. Fetal resorption was macroscopically defined by diminished fetoplacental tissue size. Histological observation demonstrated the necrotic remnants of the fetus and a dramatic reduction in placental size at the sites of the fetal loss (Figure 1B).

As shown in Figure 2A, an expected fetal resorption rate of 74% ($n = 11$) was found in the control group that was not exposed to TSS in the diet. A significant reduction in the fetal resorption was observed in the TSS-fed group (38%, $n = 9$, $P = 0.029$). The weights of surviving fetoplacental units were comparable between the TSS-fed group and the control group (Figure 2B).

3.2 | The effects of TSS on AGC-induced cytokine secretion into the peripheral blood

Next, we investigated the effects of TSS on peripheral cytokine levels following iNKT cell stimulation by AGC to explore potential immune

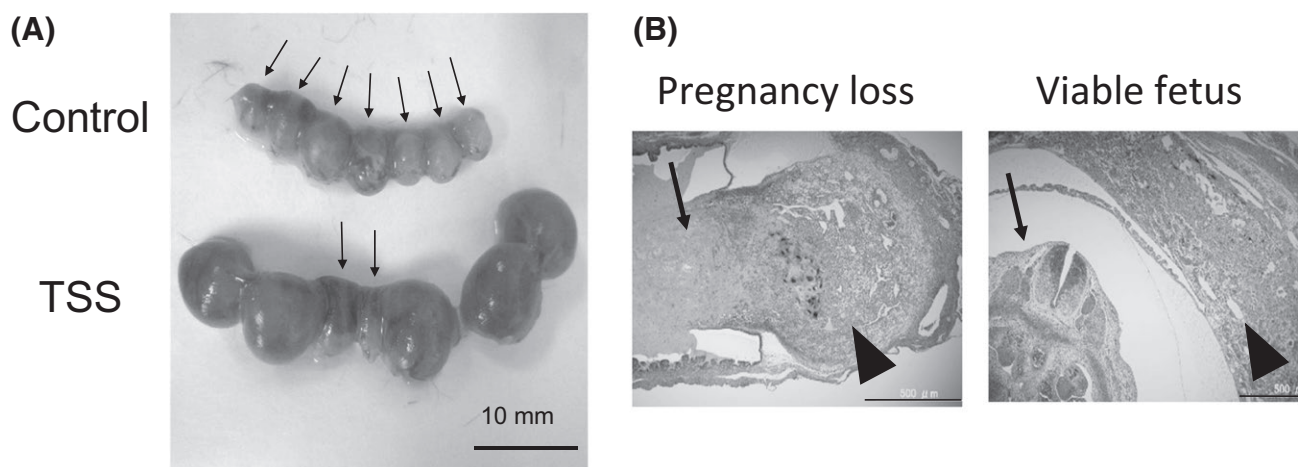


FIGURE 1 Pregnancy Loss After AGC Administration. Pregnant mice were fed a regular diet supplemented by TSS or a control diet without TSS from day 0.5 pc. Pregnant mice were injected intraperitoneally with AGC at a dose of 40 $\mu\text{g}/\text{kg}$ on day 9.5 pc. Uteri were collected at 72 h after AGC administration and pregnancy loss evaluated. A, Macroscopic images of representative pregnant uteri from control (upper) and TSS-fed dams (lower). Pregnancy loss sites are indicated by arrows. B, Microscopic findings of pregnancy loss sites. Uteri containing fetuses were fixed and were sectioned. Sections at representative sites of pregnancy loss (left) and live fetuses (right) were stained with hematoxylin and eosin. Fetal tissue is indicated by arrows and placental tissues by arrowheads

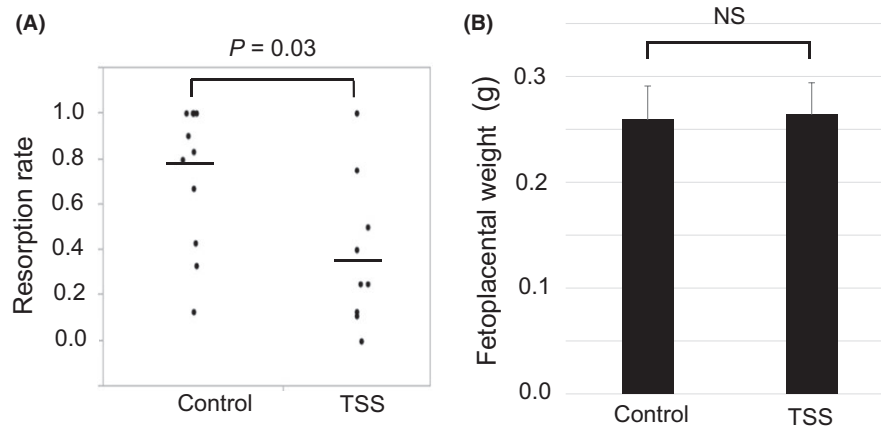


FIGURE 2 Impact of TSS Feeding on Rates of Pregnancy Loss and Fetoplacental Weights. Pregnant mice were exposed to TSS-supplemented ($n = 9$) or control (Control, $n = 11$) diet from day 0.5 pc. AGC was injected intraperitoneally in all animals to stimulate iNKT cells on injection on day 9.5 pc. The incidence of fetal absorption was evaluated 72 h later on day 12.5 pc. A, Pregnancy loss rates. The rate of pregnancy loss after AGC administration in each mouse was defined as the number of resorbed embryos divided by the number of total embryos, multiplied by 100, and expressed as a percentage. Each black dot represents a single animal. B, Fetoplacental weights in non-resorbed embryos. The fetoplacental units from pregnancies that had survived AGC injection were collected from the uteri. Fetoplacental units were weighed and averaged for the TSS-fed and control groups. Averages were graphed for comparison. NS, not significant. Error bars represent standard deviations

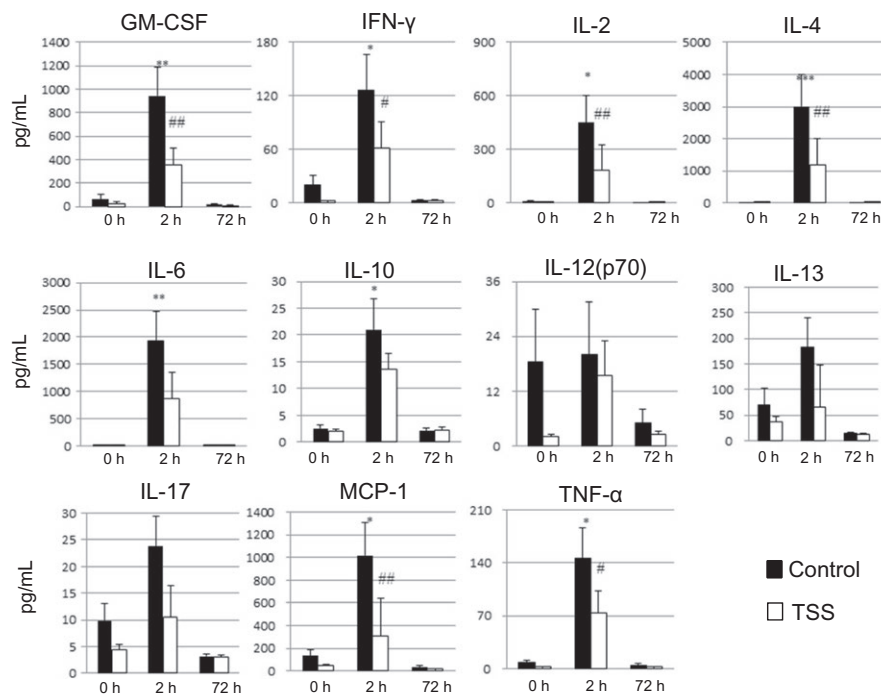


FIGURE 3 Cytokine Induction after AGC Administration. AGC was injected intraperitoneally into TSS-fed and control mice on day 9.5 pc. Serum was collected just before AGC injection (0 h) and at 2 h and 72 h after injection from the same mice. Blood cytokine levels were measured using cytokine arrays. Solid bars represent average cytokine levels in the control group and empty bars blood cytokine levels in the TSS-fed group. *: $P < 0.05$, **: $P < 0.01$, and ***: $P < 0.001$ refer to comparisons between 0 and 2 h in the control group. #: $P < 0.05$ and ##: $P < 0.01$ refer to comparisons between the control and TSS-fed groups at 2 h

changes associated with reductions in pregnancy loss (Figure 3). In previous works, cytokine release in response to iNKT cell activation by AGC could be detected within a few hours.^{13,15} Therefore, serum was collected from the tail veins of the dams just prior to AGC

administration and 2 hours later to detect rapid cytokine responses to iNKT activation. An additional serum sampling was performed at 72 hours post-AGC administration at the time of killing to evaluate pregnancy loss in the same mice. In control mice, a significant

elevation in the peripheral serum levels of GM-CSF, IFN- γ , IL-2, IL-4, IL-6, IL-10, MCP-1 and TNF- α could be detected just 2 hours after AGC administration, while levels of IL-12(p70), IL-13 and IL-17 were relatively unaffected. By 72 hours, those cytokine surges that were seen at 2 hours post-AGC administration were no longer detectable and the concentrations of these cytokines had returned to levels comparable to those of the original status before stimulation. Importantly, a remarkable suppression of AGC-associated cytokine release was measured at the 2-hour time point in the TSS-exposed group. When compared to AGC-stimulated controls that were not exposed to TSS, we detected significantly lower serum levels of GM-CSF, IFN- γ , IL-2, IL-4, MCP-1 and TNF- α in TSS-fed mice stimulated by AGC to induce iNKT cell-mediated pregnancy loss. A similar trend was observed for IL-6, IL-10, IL-13 and IL-17, although the differences between the TSS-fed group and the controls did not reach statistical significance.

3.3 | Alterations in splenic iNKT cells upon exposure to a diet containing TSS

To further examine mechanisms underlying our observed reduction in AGC-induced cytokine response by TSS-exposed pregnant

mice, we characterized the frequency of iNKT cells in the spleens of TSS-exposed and non-exposed animals. The spleens of the TSS-fed and control groups were collected on day 7.5 pc and day 9.5 pc prior to AGC administration. The percentage of iNKT cells among total splenic lymphocytes was measured using flow cytometric analyses. Splenic iNKT cells were identified as those TCR β -positive cells that also bound the CD1d tetramer (Figure 4A). As shown in Figure 4B, the ratio of iNKT cells to total splenic lymphocytes was lower in the TSS-fed mice than in the controls ($P = 0.004$) at day 7.5 pc. This pattern continued thru day 9.5 pc, although the difference between the two groups was not statistically significant ($P = 0.09$).

4 | DISCUSSION

In the present study, TSS administration protected pregnant mice from AGC-induced iNKT cell-mediated pregnancy loss. The surge in peripheral cytokine levels noted in control mice after iNKT activation by AGC was suppressed in TSS-fed mice. Dietary exposure of pregnant mice to TSS led to a reduction in iNKT cell ratio in splenic lymphocytes.

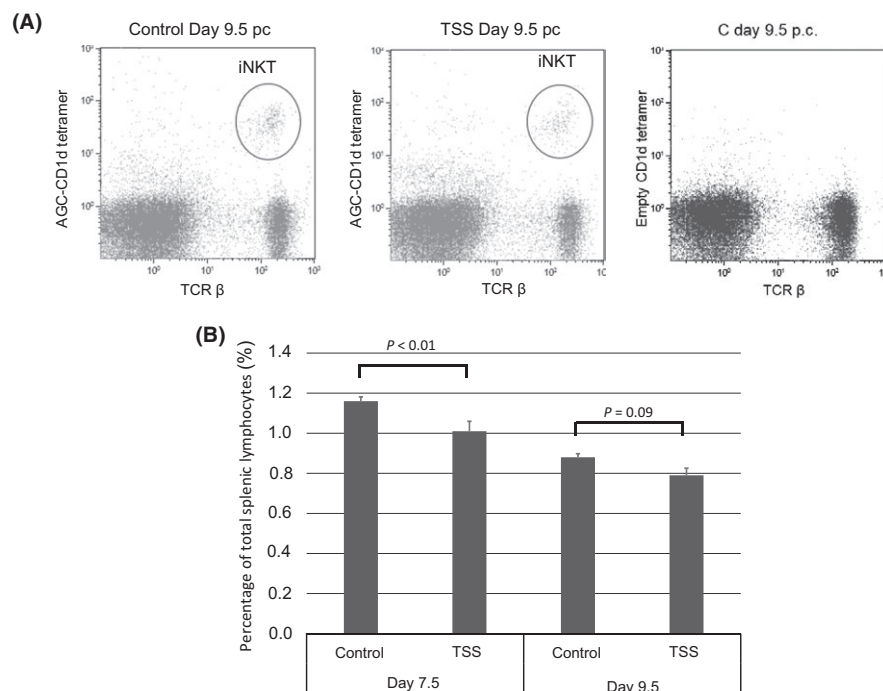


FIGURE 4 Splenic Lymphocyte Characteristics in AGC-exposed TSS-fed and Control Mice on Post-coital Days 7.5 or 9.5. Spleens were harvested from TSS-fed and control mice on day 7.5 pc or 9.5 pc. The frequency of iNKT cells among total splenic lymphocytes was measured using flow cytometry as described in the "Materials and Methods". A, Identification of iNKT cells using flow cytometry. iNKT cells were defined as TCR β -positive and AGC-CD1d tetramer-positive. A representative plot for splenic lymphocytes obtained from control (left) and TSS-fed (middle) groups on day 9.5 pc is shown. Staining result of control feeding group using empty CD1d tetramer without AGC loading is shown as a negative control (right). The circles and the percentages demonstrate iNKT cell subsets and their frequency among the total splenic lymphocytes (left and middle). B, Percentage of iNKT cells among splenic lymphocytes in TSS-fed and control animals. The ratios of iNKT cells among total splenic lymphocytes were determined by flow cytometry. Comparisons are shown for control ($n = 8$ for both day 7.5 pc and day 9.5 pc) and TSS-fed groups ($n = 8$ for both day 7.5 pc and day 9.5 pc). Error bars represent standard errors. P values were determined using student t test analyses

Invariant natural killer T cells are characterized by unique activation mechanisms. Using an "invariant" T-cell receptor, iNKT cells recognize glycolipid antigen present on CD1d by antigen-presenting cells and immediately release abundant amounts of a wide array of cytokines.⁸ Consistent with our findings in this study, Sullivan et al¹⁵ previously demonstrated a time-limited elevation in peripheral cytokine levels caused by their direct release from iNKT cells beginning at 1.5–2 hours after AGC administration in mice. In related work, Boyson et al¹² demonstrated that AGC administration induced a high cytokine response and pregnancy loss in C57BL/6 J mice, while BALB/cJ mice exhibited poor cytokine response to AGC and negligible pregnancy loss. Together, these studies suggest that the pregnancy loss that results from iNKT cell activation at day 9.5 pc is cytokine mediated. In this context, the mitigation of AGC-induced pregnancy loss in TSS-fed mice in this study is likely related to their tempered cytokine response to AGC exposure.

Accumulating evidence supports a critical involvement of iNKT cells in the immunological regulation of successful pregnancies.¹⁶ In human pregnancy, iNKT cells residing in the decidua are 10 times more frequent than those found in peripheral blood.¹¹ These decidual iNKT cells are uniquely characterized by an abundant production of GM-CSF.¹¹ Excessive activation of iNKT cells is associated with adverse pregnancy outcomes in mice, including increased pregnancy loss in early pregnancy and preterm birth later in pregnancy,^{12,17} strongly suggesting that fine tuning of iNKT cell activity is important to pregnancy success.² Despite protean efforts to identify causative factors, about a half of RPL cases remain unexplained. Dysregulation of cytokine environment at the maternal-fetal interface, especially toward one that is Th1 and Th17 biased, could be causative in some of these cases of unexplained RPL.¹⁸ Although it continues to remain unclear whether disruption of the physiologic activities of iNKT cells is related to the pathology of RPL, an intriguing hypothesis based on our observation in the mouse model is that TSS could protect some women with RPL from pregnancy loss by modulating iNKT cell activities and promoting an appropriate cytokine milieu at the maternal-fetal interface.

Tokishakuyakusan is a Japanese herbal, "Kampo," medicine commonly used for the treatment of women with a history of adverse pregnancy outcomes, including RPL.¹⁹ The clinical usage of TSS is empirical, and the molecular mechanisms underlying the purported efficacy of TSS are limited. Regarding biological effects of TSS, vascular smooth muscle relaxation,²⁰ inhibition of platelet aggregation²¹ and anti-inflammatory effect including induction of regulatory T cell^{22,23} have been reported previously. This study is the first to report a potential mechanism to explain a documented improvement in pregnancy maintenance in mice upon dietary exposure to TSS. Using a murine model of pregnancy loss induced by iNKT stimulation by AGC, we found that dietary TSS diminished the magnitude of the AGC-induced cytokine storm. As reductions were observed in diverse types of cytokines with different biological activities, our results cannot identify those key cytokines involved directly in improving pregnancy maintenance after exposure to dietary TSS. To this point, Ito et al¹⁰ reported that AGC-mediated pregnancy loss

was abolished in both IFN- γ - and TNF- α -deficient mice as well as in mice lacking iNKT cells secondary to a deletion of the iNKT-specific T-cell receptor component, V α 14. They proposed that the induction of IFN- γ and TNF- α is mechanistically essential to pregnancy loss after AGC stimulation of iNKT cells.¹⁰ In our previous work, pregnancy loss did not occur after the administration of OCH, another iNKT cell-specific stimulant. In contrast to AGC, the cytokines induced after OCH stimulation of iNKT cells are less diverse and biased toward Th2 polarity.²⁴ In the same study, supplementation with IL-4 inhibited pregnancy loss caused by iNKT cell activation with AGC.²⁴ Taken together, we hypothesize that the diminished production of IFN- γ and TNF- α and the suppression of a Th1-biased cytokine environment may explain the TSS-mediated inhibition of pregnancy loss reported in this study.

Our previous work reported that TSS capacity to induce Th-1-polarized cytokine production from mononuclear cells was associated with therapeutic potential for RPL caused by autoimmune disorders.⁶ Our previous findings are seemingly contradictory to the above-mentioned speculation that the suppression against harmful effect of Th1 cytokines by Toki contributes to the prevention of fetal resorption in this study. This might be attributable to the difference in experimental settings between the *in vitro* cell response in our previous work and the systemic outcome in mouse model. It is also possible that therapeutic effect of TSS can cover various types of immunological abnormality in RPL by compensating cytokine milieu imbalance regardless of Th1/Th2 polarity.

Lee et al²⁵ recently demonstrated that iNKT cells are abundant in the red pulp regions of the spleen. Further, intravenous injection of AGC induced iNKT activation in the spleen, but not in the thymus or lymph nodes. In our study, we found a reduction in the frequency of iNKT cells in the spleen in TSS-fed pregnant mice when compared to non-exposed pregnant controls. Additionally, TSS administration diminished the peripheral cytokine surge following AGC activation of iNKT cells. Although it was not examined whether the TSS-induced reduction in splenic iNKT cells was prolonged after AGC administration, some suppressive effects by TSS on iNKT cell activities preceding AGC stimulation might result in the observed diminished cytokine release from iNKT cells after the stimulation, leading to the inhibition of fetal resorption. A previous study described that regulatory T cells induced by TSS improved the survival of allogeneic cardiac grafts in mice.²² The suppression of iNKT activities in this study might result from such enhanced regulatory T-cell activities by TSS. Supporting this idea, a recent study by another group reported that AGC-induced pregnancy loss was inhibited by adoptive transfer of regulatory T cells.¹⁷

Invariant natural killer T cells will become unresponsive to stimulation or anergic following repetitive stimulation.²⁶ This phenomenon can limit the clinical application of iNKT stimulants for cancer therapy. It is unclear from our study whether TSS administration could also lead to anergy as TSS is a complex mixture of herbal constituents and a glycolipid antigen contained in any single ingredient or a combination of ingredients may be responsible for the effects seen. Interestingly, recent studies propose the new concept that glycolipid

factors derived from bacteria colonizing medicinal plants contribute to the immune-boosting activities.^{27,28} In fact, plant-colonizing microorganisms such as *Rahnella* species have been linked to the immunostimulatory effects of *Angelicae Radix*, a central ingredient of TSS.²⁸ In this context, the molecular interaction of bacteria-derived glycolipid components in TSS with iNKT cells could be a compelling target for future investigation.

We need to be cautious about the interpretation of the findings in this study. We did not fully prove that the diminished cytokine response by TSS in the peripheral blood was attributable solely to the altered activities of splenic iNKT cells. A previous study¹¹ demonstrated that iNKT cells are highly accumulated in the uterine endometrium during pregnancy. It is possible that modified activities of uterine iNKT cells were involved in the inhibitory effect of TSS on fetal resorption observed in this study. These issues need to be investigated in the future.

In summary, our findings propose a new mechanism by which a commonly used Kampo medicine could affect human disease-immune modulation of iNKT cells. The knowledge gained in this study may contribute to better understanding of the pathoetiology of RPL and could ultimately provide a scientific basis for RPL therapy using TSS.

CONFLICT OF INTEREST

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